Experiment No 1 & Case Study 1

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1. Experiment / Case Study No : 1

2. Experiment Title:

Demonstrate Inline, Internal and External JavaScript; use console methods ; create a webpage showing user info and a welcome webpage.

3. Software / Tools Required:

- Text editor — VS Code
- Web Browser (Google Chrome / Mozilla Firefox / Microsoft Edge) with built-in Developer Console
- Basic knowledge of HTML to embed JavaScript using the <script> tag

4. Experiment Program Code:

inlinejs.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Document</title>
</head>
<body>
   <h2>Inline Javascript Example</h2>
   <button onclick="alert('Welcome to Javascript!')">Click me</button>
</body>
</html>
```

internaljs.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Internal Javascript</title>
</head>
<body>
   <h2>Internal Javascript Example</h2>
   <p id="demo"></p>
   <script>
      document.getElementById("demo").innerHTML= "Welcome to Internal js";
   </script>
</body>
</html>
```

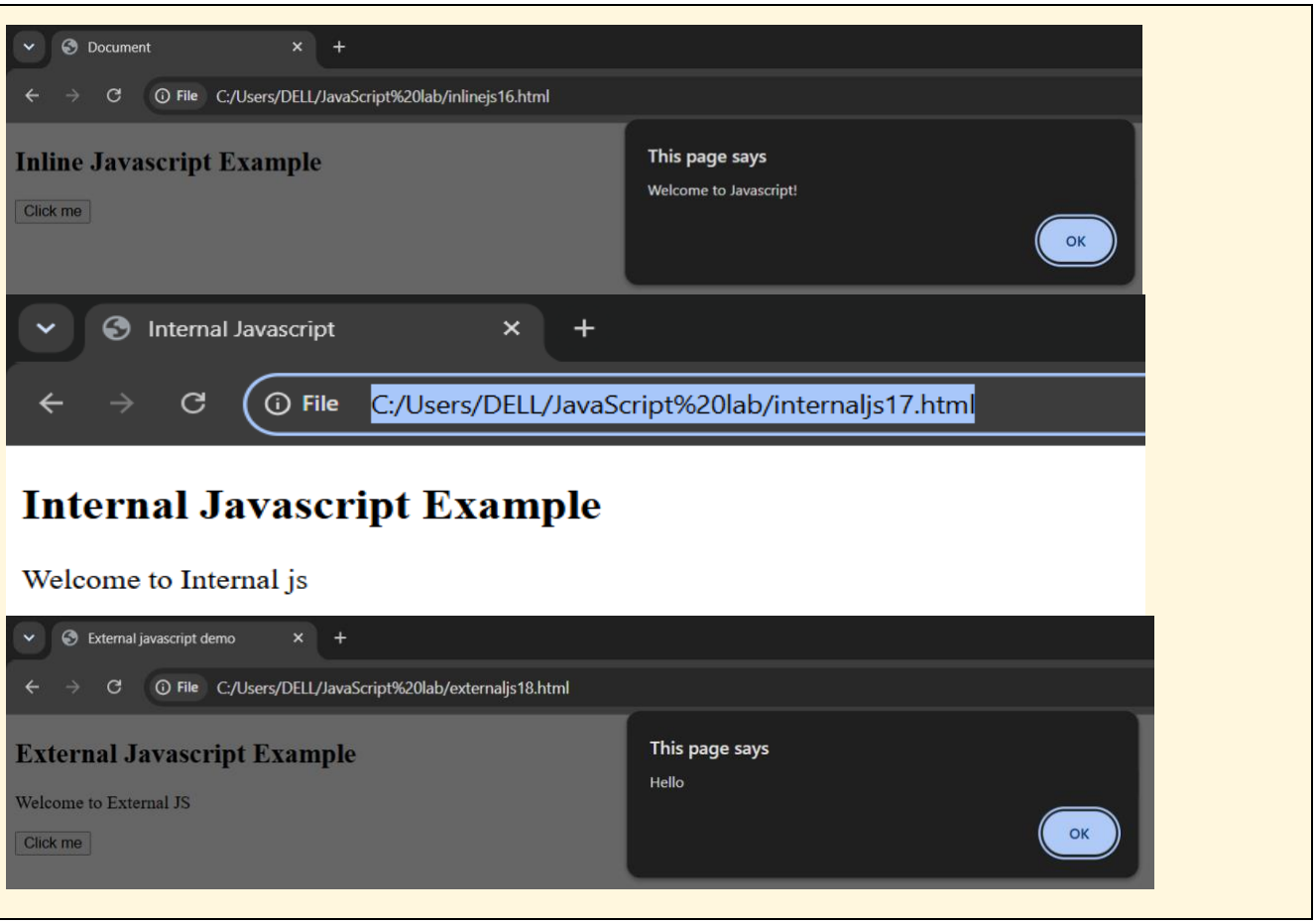
externaljs.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>External javascript demo</title>
</head>
<body>
    <h2>External Javascript Example</h2>
    <p id="demo">Welcome to External JS</p>
    <button onclick="changeText()">Click me</button>
    <script src="script18.js"></script>
</body>
</html>
```

script.js file

```
function changeText() {
    document.getElementById("demo").innerHTML = "Hello ! This text";
    window.alert("Hello");
}
```

4. Output:



5. Case Study Title:

SIT NAGPUR WEBPAGE

Case Study Background

JavaScript can be classified in two common ways in academic practicals: (a) by where the script code is written and placed in a document Inline, Internal, and External JavaScript, and (b) by where the script executes Client-side JavaScript and Server-side JavaScript. This case study demonstrates external way of writing JavaScript with code.

A. Ways of Writing JavaScript

- Inline JavaScript — code written directly inside an HTML tag using an event attribute, e.g. onclick.
- Internal JavaScript — code written inside a <script> tag within the same HTML file, usually in the <head> or before </body>.
- External JavaScript — code written in a separate .js file and linked to the HTML file using the src attribute.

B. Client-side vs Server-side JavaScript

- Client-side JavaScript runs in the user's web browser (e.g. form validation, DOM manipulation, animations). It does not need a server to execute.
- Server-side JavaScript runs on a server using a runtime such as Node.js (e.g. handling database queries, authentication, and API requests) before sending results to the browser.

6. Case Study Program Code:

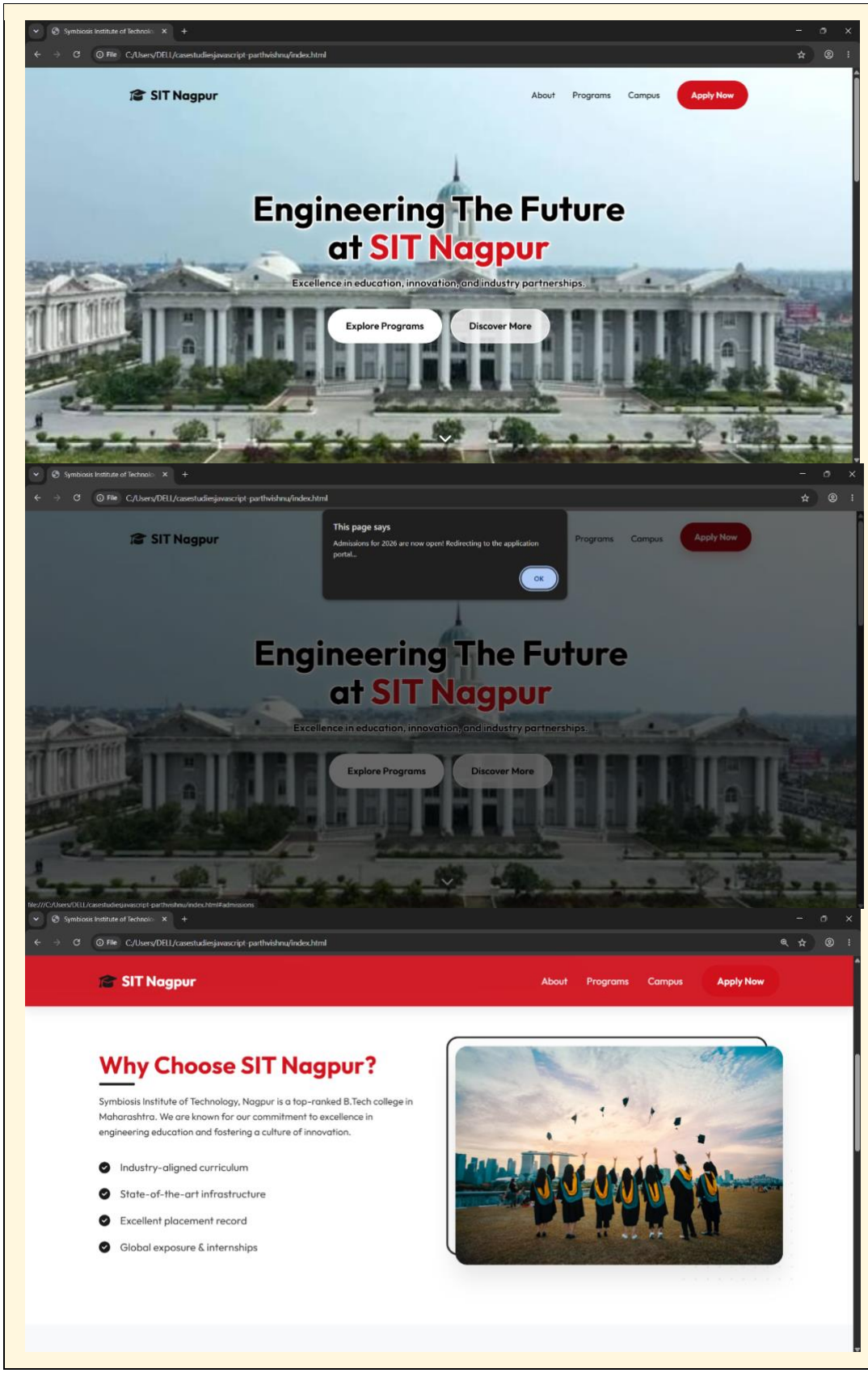
style.css

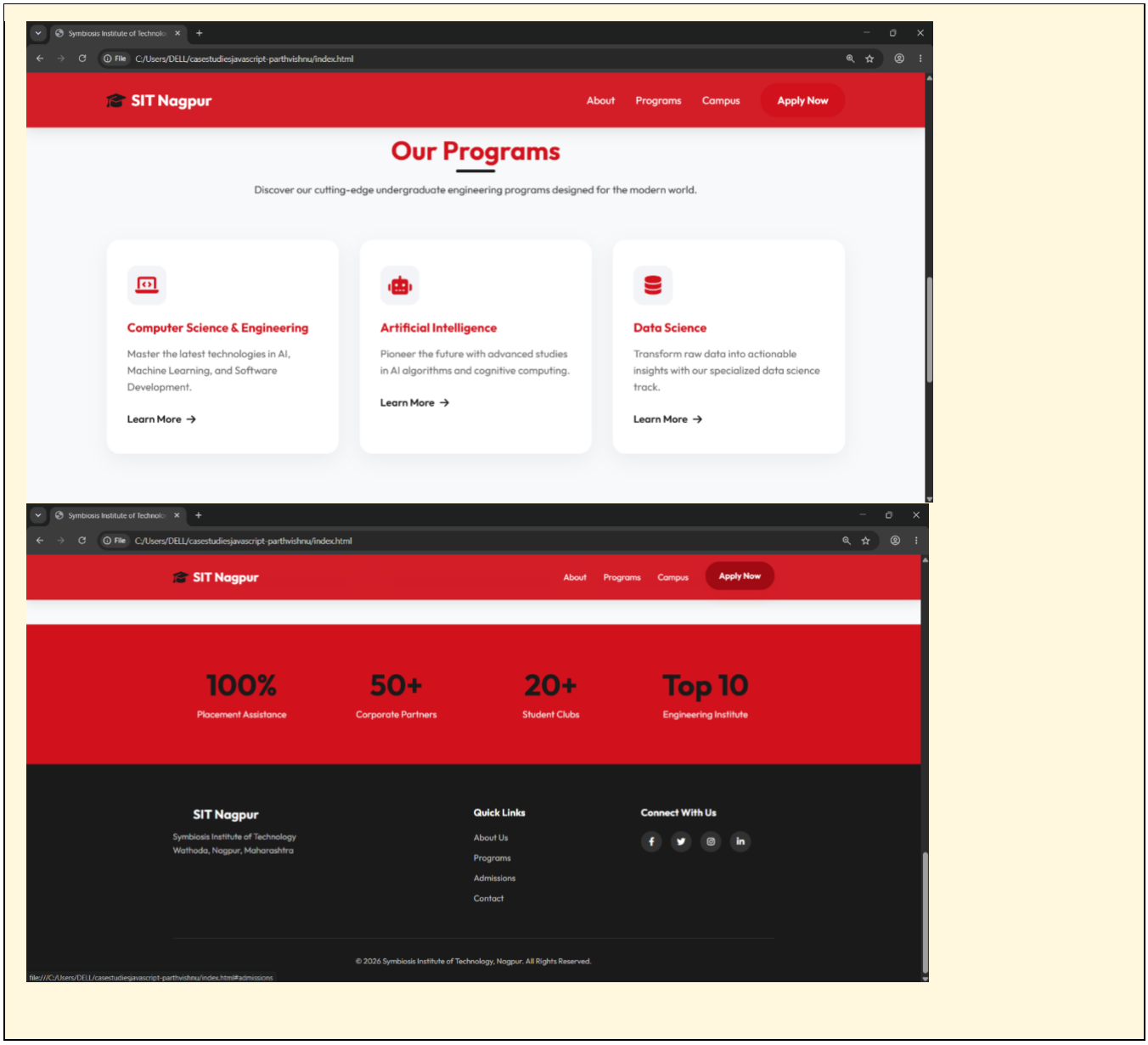
script.js file

index.html file (linking the external script)

codes uploaded on github in the same folder

7. Output:


SIT Nagpur
About
Programs
Campus
Apply Now
Engineering The Future
at SIT Nagpur
Excellence in education, innovation, and industry partnerships.
Explore Programs
Discover More
This page says
Admissions for 2026 are now open! Redirecting to the application portal...
OK
Why Choose SIT Nagpur?
Symbiosis Institute of Technology, Nagpur is a top-ranked B.Tech college in Maharashtra. We are known for our commitment to excellence in engineering education and fostering a culture of innovation.
Industry-aligned curriculum
State-of-the-art infrastructure
Excellent placement record
Global exposure & internships



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8. Result / Conclusion:

Thus, the different data types in JavaScript primitive types (Number, String, Boolean, Undefined, Null, Symbol, BigInt) and non-primitive types (Object, Array, Function) were studied and implemented successfully using the typeof operator. Additionally, the three ways of writing JavaScript Inline, Internal, and External along with the distinction between Client-side and Server-side JavaScript were studied and demonstrated with working examples. It was observed that External JavaScript is the most preferred method for larger projects as it keeps HTML and JavaScript code separate, improving readability and reusability.